

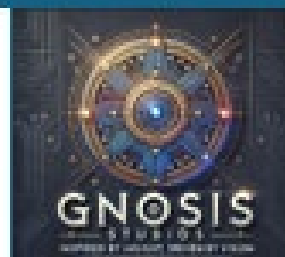


QUANTUM SLIP

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Quantum Slip is a stealth-based game inspired by genre-defining titles such as Metal Gear Solid, Splinter Cell, Thief, and Dishonored. These games expertly create tension through mechanics that demand strategic thinking, environmental awareness, and the constant risk of detection. The thrill of carefully navigating enemy patrols, planning movements with precision, and experiencing the adrenaline rush of near discovery has always fascinated me (McKeand, 2022).

Building on the knowledge gained from my coursework and utilising university-provided resources, I set out to design and develop a vertical slice of a level for Quantum Slip. This project is both a creative and technical endeavour, allowing me to deepen my understanding of stealth mechanics, level design principles, and player engagement. My goal is to capture the core elements that make stealth gameplay so compelling while crafting an immersive and cohesive game environment (Rollings and Morris, 2004).

Throughout the process of developing this level, I encountered numerous obstacles that necessitated problem-solving and active learning. One of the most significant challenges was mastering Unity's animator and ensuring that animation parameters responded correctly to player inputs. Initially, I experimented with blend trees, which facilitate smooth transitions between walking, jogging, and running animations. However, due to time constraints and my current skill level, I opted for a simpler approach, using Boolean values to toggle animations on and off (McShaffry, 2005).

I sourced my character models and animations from Mixamo.com (Adobe, n.d.). Utilising pre-made assets proved to be a substantial time-saver and an effective way to practice game development on smaller projects. Nevertheless, I recognise that in future projects, I will need to develop the ability to sculpt and create my own digital assets (Stabile, Prior and Hinkes, 2020).

One animation from the Adventure Pack, the "wall hug," particularly stood out to me. As a fan of the stealth genre, I was eager to implement it as a functional mechanic (Mixamo, no date).

The Wall Hug script required the creation of a mask layer named "wall" to ensure the player could perform the animation in the correct position

without physics or gravity interfering. This proved to be more challenging than anticipated. I also attempted an early implementation of side-scrolling along the wall, but my limited knowledge of physics interactions prevented it from functioning as intended. To resolve this, I introduced a cooldown function in the script to prevent animation loops and froze the player's model in place while the wall hug boolean was active. Although this solution was not perfect, it was the best I could achieve with my current understanding, and I was pleased that it functioned relatively close to my initial vision.

Next, I tackled enemy AI and patrol behaviour. Unity's NavMesh proved to be an invaluable tool, simplifying AI movement and tracking (Bennett and Sagmiller, 2014). After dedicating considerable time to animating the player character, I found the process of animating enemy characters much more straightforward. Notably, I observed an improvement in my skills, which evoked both excitement and apprehension; excitement at bringing my ideas to life and apprehension regarding the steep learning curve of digital creation. I reminded myself to remain inquisitive and embrace the learning process (Ambrose et al., 2010).

For the AI, I wanted enemies to have both vision and audio detection capabilities. I experimented with multiple script iterations and consulted university tutorials, eventually developing a basic enemy AI with predetermined patrol locations (Blackboard, no date). Balancing patrol and chase speeds was crucial to maintaining an engaging gameplay experience. I also began work on a shadow detection system, allowing shadows to function as hiding spots for the player. Given that I used multiple spotlights instead of a single directional light, implementing this mechanic proved exceptionally difficult. I ultimately scripted the system to detect light sources dynamically rather than relying on predefined baked lighting. While this approach functioned to some extent, it introduced conflicts with other mechanics, making my code feel precarious, like a Jenga tower. Given more time, I would have completely rewritten the AI and stealth system, but for the scope of this project, a functioning version had to suffice.

The teleportation mechanic, surprisingly, was relatively straightforward to implement. Having previously experimented with teleportation scripts, I was already familiar with the process, as teleportation is one of my

favourite mechanics in both games and storytelling (Mitchell and Vosmeer, 2021). Once I had a working player and AI system, I used assets from the Unity Asset Store to construct the facility's outer perimeter, which serve as the first level. I had initially expected this task to be tedious, yet I found it more enjoyable than anticipated. I am learning that not all aspects of digital creation are glamorous; some demand sheer dedication and hours of work to produce something worthwhile (Millerd, 2024).

Did I mention the testing? The endless testing? I experimented with various objects from an industrial asset pack, designing treacherous paths and hidden corners. Having enjoyed games like Cities: Skylines and SimCity, I found the level design aspect particularly engaging (Bereitschaft, 2015).

Lighting was another major focus, yet despite my best efforts, it remained imperfect. I now understand why entire departments are dedicated to light rendering and physics in professional game development. Nevertheless, I gained valuable experience with Unity's lighting tools and look forward to improving in future projects where lighting will play a more central role (Katatikarn and Tanzillo, 2016).

Deciding on the number of enemy guards in the level was a delicate balancing act. Too many or too few would impact the overall enjoyment, and given the unpredictability of my AI, achieving the right balance was difficult. I created multiple parented guard groups with varied patrol points, testing different configurations until the level felt suitably challenging. Again, did I mention the testing? (Schultz and Bryant, 2016).

Scene transitions turned out to be simpler than expected, reinforcing my growing understanding of Unity's system. Once the playable level was functional (though still requiring refinement), I shifted focus to the splash screen, main menu, and end screen. To avoid burnout from continuously working on the main level, this provided a refreshing change of pace. Rather than creating a brand-new scene for the main menu, I followed my lecturer's advice and reused assets from my level. I duplicated the scene, removed unnecessary elements, adjusted the camera to a cinematic angle, and prefabbed enemy models to move between two

points without interfering with the menu. I was quite pleased with the final result.

Seeing my level from a first-person perspective, one the player would not typically experience due to the top-down camera, gave me the idea to include a small cutscene at the start. Inspired by cinematic storytelling, I animated the camera along a path using Unity's Timeline, incorporating keyframes for smooth transitions and motion pauses for dramatic effect. Initially, I included exposition text during this sequence, but following feedback from play testers, I reduced the text for improved readability and planned to replace it with voice-over narration (Oliver et al., 2015).

I also experimented with a cinematic game logo and opening credits. Was it excessive? Possibly. But I thoroughly enjoyed the creative process.

Following further feedback from my lecturer, I incorporated a small tutorial area where players could familiarise themselves with the controls and mechanics. I created typewriter-style text displays, timing them to guide the player smoothly, and, once again, tested extensively. Additionally, I saved a separate version of the playable level to ensure that if the player was captured and the scene reloaded, only basic controls and objectives were displayed, rather than the initial tutorial text.

The end screen was initially just a formality, but I then had the idea to turn it into a behind-the-scenes showcase. I included screenshots documenting my development journey, adding a personal touch. You will have to complete the demo of Quantum Slip to see development screenshots.

A guest speaker named Paul, from a digital industry company, visited our class and offered an insightful suggestion, one that was so obvious I was astonished I had overlooked it. "How does the player quit the game or return to the main menu? Control + Alt + Delete?" Of course! I had neglected one of the most fundamental aspects of user interface design. Sometimes, insight lies in knowing when to keep things simple and when to introduce complexity. A very helpful reminder indeed (Maeda, 2006).

I also sought advice on how to determine when a project is finished. I find myself constantly wanting to add features, tweak mechanics, or even rewrite entire systems. In a corporate environment, deadlines and budgets dictate the conclusion of development, but what about passion projects? I suspect this is a question I will answer through experience (Cameron, 2020).

I spent my final hours squashing bugs, fixing floating objects, adjusting AI behaviour, and refining the overall experience as much as time allowed. Although I was unable to rewrite everything, I am proud of what I accomplished. This project was an invaluable learning experience, and I am excited to continue honing my skills as I progress in game development.

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